

JULAIA PORT, Kuwait

WMO: 405930

Lat: 28.867N

Lon: 48.283E

Elev: 9

StdP: 101.22

Time Zone: 3.00 (E03)

Period: 07-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.4	8.9	-5.2	2.4	10.0	-3.0	2.9	11.5	21.2	17.1	19.5	15.3	5.9	300	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	40.6	23.5	39.6	23.7	38.8	24.0	31.4	34.7	30.8	34.5	30.2	34.5	7.9	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.6	28.3	34.3	29.9	27.1	34.1	29.1	25.8	34.0	106.1	34.8	103.0	34.4	99.8	34.3	36.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	15.3	12.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	14.1	15.7	19.6	24.4	30.2	34.6	35.9	35.6	33.3	28.7	21.4	16.2
	DBStd	8.22	2.86	2.61	2.35	2.37	2.32	1.69	1.67	1.50	1.51	1.87	2.86	2.65
	HDD10.0	4	4	1	0	0	0	0	0	0	0	0	0	0
	HDD18.3	307	134	78	13	0	0	0	0	0	0	7	75	
	CDD10.0	5788	131	160	297	432	626	738	803	793	698	579	341	192
	CDD18.3	3049	2	4	52	182	367	488	544	534	448	321	97	8
	CDH23.3	44581	9	1	74	1088	4962	8069	9242	9095	7329	4212	498	2
	CDH26.7	28758	4	0	15	180	2570	5669	6762	6615	4931	1949	64	0
Wind	WSAvg	5.8	6.0	6.1	6.2	4.9	5.8	6.6	5.6	5.4	5.1	5.5	6.1	5.8
Precipitation	PrecAvg	182	35	23	31	19	5	1	1	1	0	4	32	30
	PrecMax	298	103	81	97	52	20	2	3	2	1	26	182	90
	PrecMin	82	1	1	1	1	0	0	0	0	0	1	1	
	PrecStd	61	26	20	24	15	5	1	1	0	6	39	24	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	22.0	28.9	32.2	37.3	41.1	42.1	41.6	39.6	35.9	29.6	23.1
		MCWB	27.7	16.7	17.2	19.0	21.5	22.4	22.8	24.6	24.3	23.7	21.1	17.2
	2%	DB	20.1	20.7	24.9	29.8	35.4	39.6	40.7	40.2	38.2	34.2	28.3	22.1
		MCWB	16.4	16.0	16.5	19.3	22.1	22.8	23.2	24.4	24.2	22.6	21.3	17.5
	5%	DB	19.1	19.9	23.6	28.5	34.3	38.6	39.8	39.3	37.3	33.0	27.2	21.3
		MCWB	15.5	15.5	17.4	19.7	22.2	23.2	23.4	24.8	24.4	23.6	20.4	16.9
	10%	DB	18.2	19.2	22.7	27.5	33.4	37.6	39.0	38.4	36.5	32.2	26.0	20.7
		MCWB	14.8	15.3	17.1	19.8	22.5	23.2	23.7	25.1	24.8	23.4	19.4	16.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	18.7	20.0	23.2	27.2	29.7	31.9	32.3	31.3	28.7	24.8	20.0
		MCDB	21.4	20.2	22.5	27.8	31.8	34.4	34.5	35.1	34.5	32.2	27.9	22.0
	2%	WB	17.6	17.6	19.5	22.5	26.4	28.7	30.9	31.7	30.3	27.6	23.3	19.1
		MCDB	19.2	19.4	22.2	27.0	31.5	34.3	34.4	34.8	34.2	31.6	26.7	21.4
	5%	WB	16.4	16.9	19.1	21.9	25.7	27.8	30.2	31.2	29.8	26.7	21.7	18.0
		MCDB	18.3	19.1	21.9	26.2	31.1	34.3	34.3	34.6	34.1	30.7	25.8	20.7
	10%	WB	15.3	16.2	18.6	21.3	25.0	26.8	29.5	30.7	28.9	25.9	20.5	16.9
		MCDB	17.7	18.6	21.5	25.6	30.8	34.6	34.4	34.7	33.9	30.2	24.8	20.0
Mean Daily Temperature Range	5% DB	MDBR	5.4	4.9	4.5	4.5	4.5	5.6	5.6	5.2	5.9	5.9	6.1	6.1
		MCDBR	5.2	4.7	5.7	6.1	5.8	6.9	6.9	6.5	7.1	6.8	6.9	6.1
		MCWBR	4.9	4.7	5.4	5.9	7.1	7.6	8.0	8.1	8.8	8.2	6.5	5.6
	5% WB	MCDBR	4.6	3.5	3.5	4.6	4.5	5.3	4.5	4.4	5.1	5.5	5.4	4.6
		MCWBR	5.0	4.1	4.1	5.2	6.7	9.3	8.1	7.4	7.6	7.2	5.7	5.1
Clear-Sky Solar Irradiance	taub	0.516	0.564	0.617	0.703	0.703	0.663	0.681	0.631	0.576	0.569	0.550	0.513	
	taud	1.823	1.729	1.617	1.519	1.525	1.567	1.567	1.657	1.751	1.764	1.793	1.842	
	Ebn at Noon	690	698	693	653	657	680	666	695	717	683	652	667	
	Edh at Noon	180	215	255	290	289	275	275	249	220	204	184	170	
All-Sky Solar Radiation	RadAvg	3.38	4.18	5.23	5.71	6.57	7.28	6.96	6.65	5.97	4.80	3.52	3.10	
	RadStd	0.23	0.27	0.26	0.31	0.29	0.28	0.19	0.15	0.21	0.20	0.27	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.33	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	+94	+107

Nomenclature: See separate page